

COURSE DESCRIPTION			
Program	Printing Technology		
Course Title	Digital Office Systems and Documentation		
Course Code	2109	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

Printing technologists today are expected to handle: Job documentation, Production reports, Publications, Salary and mark sheets, Client presentation, Certificates, newsletters, and formatted documents. This workshop equips students with practical office automation skills directly applicable to printing offices, academic institutions, and production environments

Course Objectives

1	Create professional documents using word processing tools- Ms word, excel, and powerpoint
2	Design newsletters, certificates, and cover pages suitable for print
3	Prepare tabulated data, mark lists, and salary sheets using spreadsheets
4	Apply formulas and formatting for data accuracy
5	Develop effective presentation slides with animations and transitions
6	Use office automation tools for real-world printing and administrative tasks

Course Pre-requisite

Topic	Course code	Course Title	Semester
Designing of letterhead	1101	Introduction to Printing Technology	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Prepare print-ready documents used in offices and printing houses	Apply
CO2	Gain competency in preparing structured data sheets commonly used in institutions and printing offices	Apply
CO3	Design and present professional slides for meetings, reviews, and client interactions	Apply
CO4	Demonstrate end-to-end office automation skills relevant to the printing industry- Integrate multiple office automation tools to develop complete documentation systems for real-world applications	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	-	2	-	3	-	-	-	1	2	1
CO2	1	1	2	-	3	-	-	-	1	1	1
CO3	-	-	2	-	2	-	-	-	2	3	1
CO4	1	2	3	1	3	1	-	1	2	2	2
Total	3	3	9	1	11	1	-	1	6	8	5
Correlation Level	1	1.5	2.25	1	2.75	1	-	1	1.5	2	1.25

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	R				CIE	ESE	Total	CIE	ESE	Total	

0	0	3	0	1.5	45	1.5				60	40	100	100
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Familiarization with Microsoft Word Interface	navigate the Microsoft Word interface, identify key tools, menus, and shortcuts, and customize the workspace for efficient document preparation	CO1	Apply	1
2	Document Creation and Formatting	Create professional documents using appropriate fonts, paragraph styles, spacing, alignment, headers, footers, and page setup as per office standards	CO1	Apply	2
3	Logo and QR code creation	design and create a professional logo and QR code	CO1	Apply	3
4	Cover Page Designing	design structured and visually balanced cover pages using text formatting, images, shapes, and themes suitable for reports, projects, and official documents	CO1	Apply	3
5	Newsletter Design	design newsletters using columns, text boxes, images, and layout tools while maintaining readability and consistency for institutional or corporate communication	CO1	Apply	3
6	Certificate Creation	design formal certificates with correct layout, typography, borders, logos, and alignment following standard documentation practices.	CO1	Apply	3
Module II					
7	Introduction to Microsoft Excel	Explain the excel interface	CO2	Apply	1
8	Data Entry, Cell Formatting and barcode creation	enter, format, and organize data using cell styles, borders, number formats, and alignment also create barcode	CO2	Apply	2
9	Mark List Preparation	prepare automated mark lists using formulas, functions (SUM, AVERAGE, IF), grading logic, and basic data validation techniques	CO2	Apply	3
10	Salary Sheet Preparation	create salary sheets incorporating calculations for gross pay, deductions, and net salary using formulas and structured data layouts	CO2	Apply	3
Module III					
11	Introduction to Microsoft PowerPoint	identify PowerPoint tools and slide structures required for creating effective presentations.	CO3	Apply	1
12	Slide creation- Slide Design and Formatting	design well-formatted slides using layouts, themes, images, charts, and visual hierarchy for clear communication	CO3	Apply	2
13	Creating a Short Technical Presentation	develop and deliver a concise technical presentation, organizing content logically and applying appropriate slide design principles	CO4	Apply	3
Module IV					
14	Integrated Document Preparation- Word+ excel (mail merge)	integrate Excel data with Word documents using mail merge to generate personalized letters, certificates, or reports efficiently.	CO4	Apply	1
15	Office Automation Case Study Preparing a complete office document set: Cover page, certificate, excel data sheet, presentation	Analyze an office workflow case study, identify documentation requirements, and apply suitable office automation tools to solve real-world problems	CO4	Apply	5

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Project 1: Academic Office Automation System

- Mark list in Excel
- Result report in Word
- Mail Merge for individual result sheets
- Summary presentation

Project 2: Printing Job Documentation System

- Job order form
- Cost calculation sheet
- Client presentation

Project 3: Salary and Payroll Documentation

- Attendance sheet
- Salary computation
- Pay slip generation using Mail Merge

Project 4: Event Documentation Package

- Invitation letter
- Participant database
- Certificate generation
- Event presentation

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Record Preparation- exp. No-1 &2	1
2	Record Preparation- exp. no-3	1
3	Record Preparation- exp. no-4	1
4	Record Preparation- exp. no-5	1
5	Record Preparation- exp. no-6	1
6	Create a four -page print-ready document with cover page, header, footer, and page numbers	3
Module II		
7	Record Preparation- exp. no-7 & 8	2
8	Record Preparation- exp. no-9	1
9	Record Preparation- exp. no-10	1
10	Prepare a personal budget or expense sheet in Excel using:formulas and percentage calculations	2.5
Module III		
11	Record Preparation- exp. no-11 & 12	1
12	Record Preparation- exp. no-13	1
13	Design a 5-slide PowerPoint presentation on "Importance of Office Automation in Printing Industry"	2
Module IV		
14	Record Preparation- exp. No- 14	1
15	Record Preparation- exp. No- 15 Study and submit a brief note on Integration of Word, Excel, and PowerPoint in office environments	3
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Microsoft Word Step by Step	Joan Lambert, Joyce Cox	Microsoft Press
2	Microsoft Excel Step by Step	Curtis Frye	Microsoft Press
3	Microsoft PowerPoint Step by Step	Joan Lambert	Microsoft Press

REFERENCE

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1	Microsoft Word Step by Step	Joan Lambert, Joyce Cox	Microsoft Press
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3	Microsoft PowerPoint Step by Step	Joan Lambert	Microsoft Press

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://support.microsoft.com/en-us/word	I
2	https://learn.microsoft.com/en-us/training/modules/get-started-with-word/	I
3	https://support.microsoft.com/en-us/office/create-newsletters-using-word-9e8a3d57-3c5f-4c1c-bc6e-9d8f9b8c4d2a	I
4	https://support.microsoft.com/en-us/office/available-number-formats-in-excel-0afe8f52-97db-41f1-b972-4c51f3a1e26d	II
5	https://learn.microsoft.com/en-us/training/modules/use-functions-excel/	II
6	https://learn.microsoft.com/en-us/training/modules/get-started-with-powerpoint/	II
7	https://support.microsoft.com/en-us/office/apply-or-change-a-slide-layout-0fc6a41-2f05-4d8e-9b9f-6c84b3c3a2d4	III

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	HIGH END DESKTOP SYSTEMS	Intel i5/ryzen 5 or higher, 16 Gb Ram, 512 SSD, 21" led monitor	15
2	UPS	10 KV	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Microsoft Excel	Latest version, educational license	15
2	Microsoft word	Latest version, educational license	15
3	Power point	Latest version, educational license	15

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to			15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5

Total	50
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Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50